```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Check if image is loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Convert BGR to RGB
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    # Laplacian Mask with Diagonal Neighbours
    laplacian_kernel = np.array([
        [1, 1, 1],
        [1, -8, 1],
        [1, 1, 1]
    ], dtype=np.float32)

    # Apply Laplacian filter
    laplacian = cv2.filter2D(
        img_rgb,
        cv2.CV_32F,
        laplacian_kernel
    )

    # Sharpen image
    sharpened = img_rgb.astype(np.float32) - laplacian

    # Keep pixel values between 0 and 255
    sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

    # Convert Laplacian output for display
    laplacian_display = cv2.convertScaleAbs(laplacian)

    # Display Images
    plt.figure(figsize=(15, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(img_rgb)
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 3, 2)
    plt.imshow(laplacian_display)
    plt.title("Laplacian Output")
    plt.axis("off")

    plt.subplot(1, 3, 3)
```

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    plt.imshow(sharpened)
    plt.title("Sharpened Image")
    plt.axis("off")

    plt.show()

    # Save output
    output_file = "Sharpened_Image.jpg"

    cv2.imwrite(
        output_file,
        cv2.cvtColor(sharpened, cv2.COLOR_RGB2BGR)
    )

    print("Image saved as Sharpened_Image.jpg")

    # Download output
    files.download(output_file)
```

Choose Files anime-girl-…llustration.jpg

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Saving anime-girl-writing-cafe-with-headphones-illustration.jpg to anime-girl-writing-cafe-with-headphones-illustration (1).jpg
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Image saved as Sharpened_Image.jpg
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